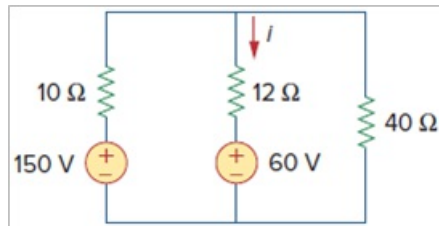


Problem

Solve for the current i in the circuit of Fig. 4.103 using Thevenin's theorem. (*Hint*: Find the Thevenin equivalent seen by the $12\text{-}\Omega$ resistor.)

Figure. 4.103:



Step-by-step solution

Step 1 of 7

Refer to Figure 4.103 in the text book.

Disconnect the $12\text{-}\Omega$ resistance to calculate the Thevenin's Voltage as shown in Figure 1.

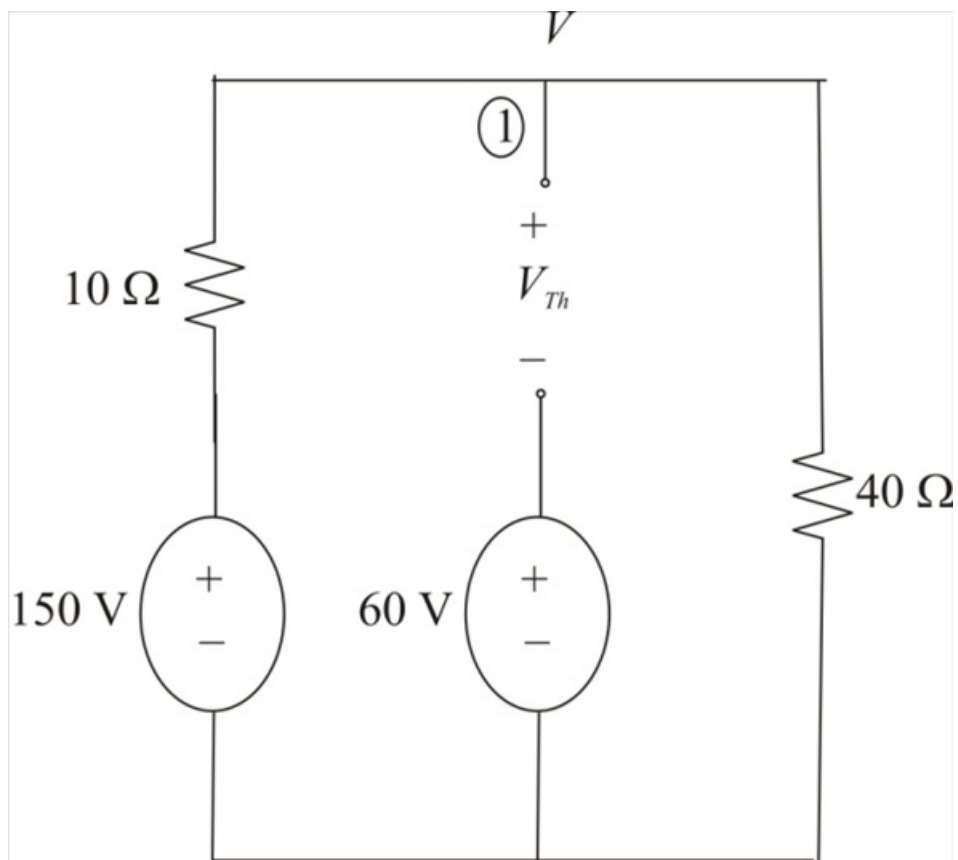


Figure 1

Step 2 of 7

Apply Kirchhoff's Current Law at node 1.

$$\frac{V - 150}{10} + \frac{V}{40} = 0$$

$$\frac{4V - 600 + V}{40} = 0$$

$$4V - 600 + V = 0$$

$$5V - 600 = 0$$

$$5V = 600$$

$$V = \frac{600}{5}$$

$$= 120 \text{ V}$$

Therefore, the value of the voltage V is 120 V .

Step 3 of 7

Calculate the value of the Thevenin's Voltage, V_{Th} .

$$V_{Th} = V - 60$$

Substitute 120 V for V in the equation.

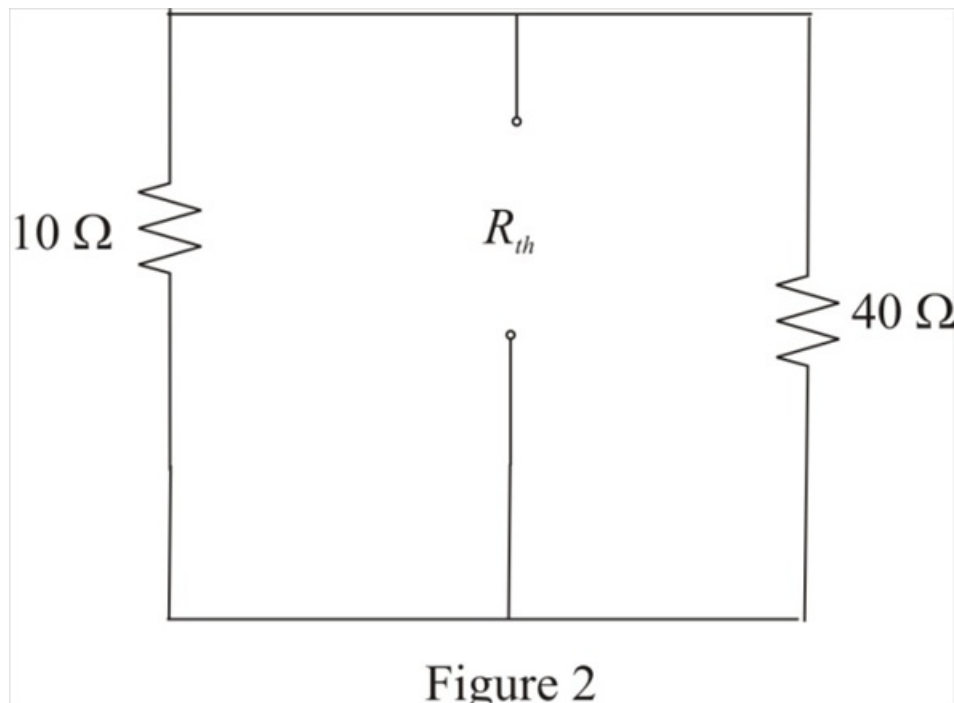
$$V_{Th} = 120 - 60$$

$$= 60 \text{ V}$$

Therefore, the value of the Thevenin's Voltage V_{Th} is 60 V .

Step 4 of 7

Deactivate the sources in the circuit diagram to calculate the Thevenin's resistance.



Step 5 of 7

Calculate the value of the Thevenin's resistance, R_{Th} .

$$\begin{aligned} R_{Th} &= 10 \parallel 40 \\ &= \frac{(10)(40)}{10 + 40} \\ &= \frac{400}{50} \\ &= 8 \, \Omega \end{aligned}$$

Therefore, the value of the Thevenin's resistance R_{Th} is $8 \, \Omega$.

Step 6 of 7

Draw the Thevenin's equivalent circuit diagram including the load resistance.

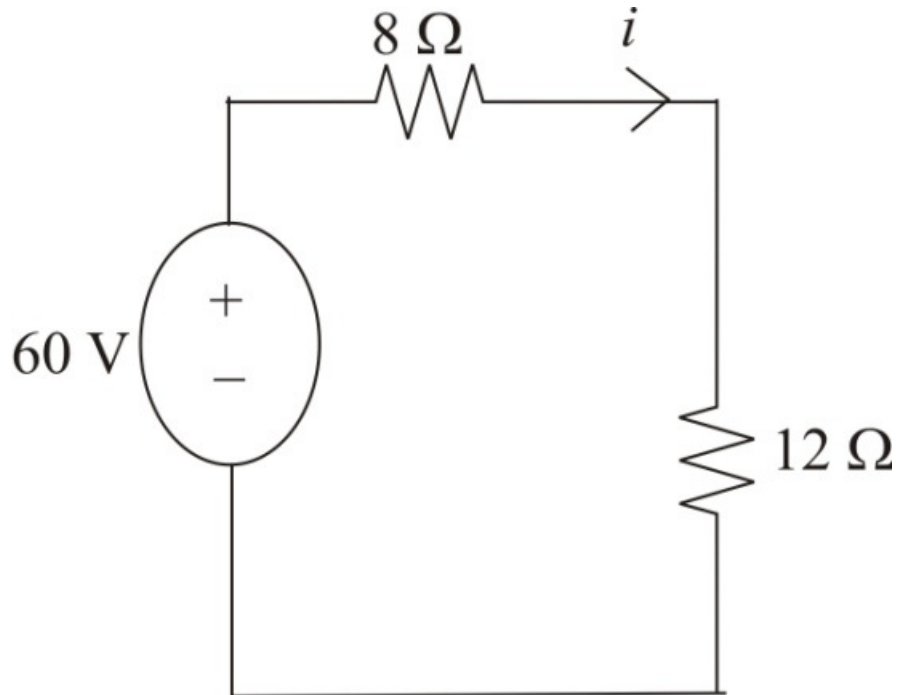


Figure 3

Step 7 of 7

Calculate the value of the current, i .

$$\begin{aligned} i &= \frac{60}{12 + 8} \\ &= \frac{60}{20} \\ &= 3 \, \text{A} \end{aligned}$$

Therefore, the value of the current i is $3 \, \text{A}$.